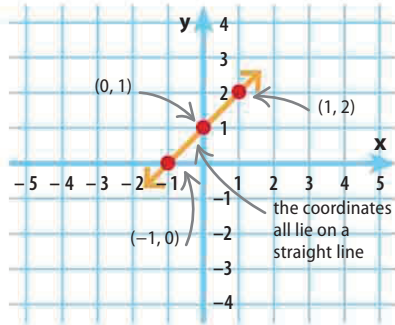


Equation of a line

Lines that pass through a set of coordinates on a pair of axes can be expressed as equations. For example, on the line of the equation $y = x + 1$, any point that lies on the line has a y coordinate that is 1 unit greater than its x coordinate.

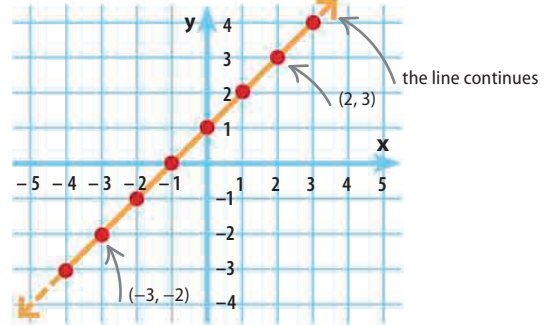


The equation of a line can be found using only a few coordinates. This line passes through the coordinates $(-1, 0)$, $(0, 1)$, and $(1, 2)$, so it is already clear what pattern the points follow.

$$y = x + 1$$

y coordinate

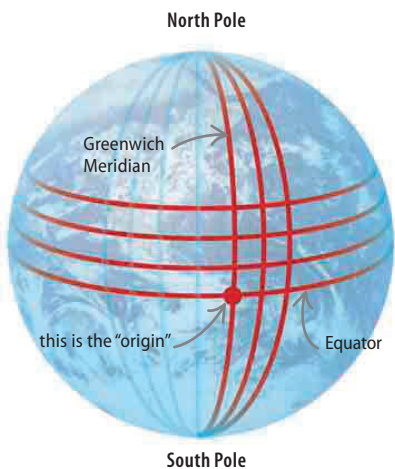
x coordinate



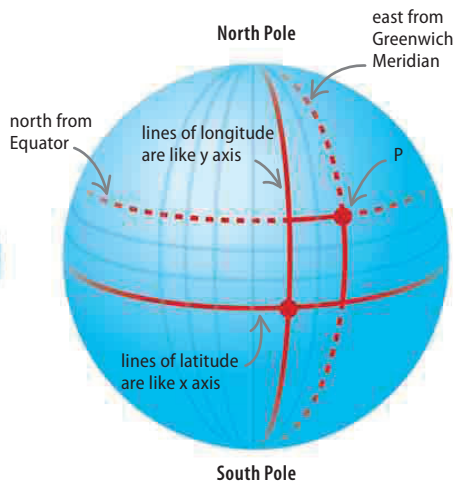
The graph of the equation is of all the points where the y coordinate is 1 greater than the x coordinate ($y = x + 1$). This means that the line can be used to find other coordinates that satisfy the equation.

World map

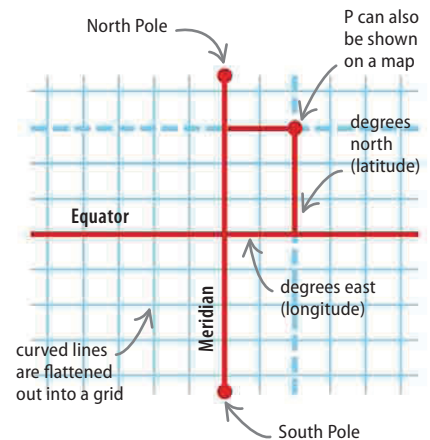
Coordinates are used to mark the position of places on the Earth's surface, using lines of latitude and longitude. These work in the same way as the x and y axes on a graph. The "origin" is the point where the Greenwich Meridian (0 for longitude) crosses the Equator (0 for latitude).



Lines of longitude run from the North Pole to the South Pole. Lines of latitude are at right-angles to lines of longitude. The origin is where the Equator (x axis) crosses the Greenwich Meridian (y axis).



The coordinates of a point such as P are found by finding how many degrees East it is from the Meridian and how many degrees North it is from the Equator.



This is how the surface of the Earth is shown on a map. The lines of latitude and longitude work as axes—the vertical lines show longitude and horizontal lines show latitude.